



Misinformation detection with automatic fact-based news verification

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ABSTRACT

Applying fact-based verification to online misinformation detection can be a complex task due to the challenges of varying input lengths and the lack of sufficient data to train fact-based verification models for misinformation detection. We collected and created two specific datasets consisting of news articles and a set of fact articles (fact pool) which are used to develop a novel online misinformation detection pipeline based on information retrieval and claim verification. In addition to the AveriTec dataset, we collected and published the Misbar¹ dataset, which consists of expert-annotated long news articles and verification articles. We also propose a novel eXplainable Misinformation Detection method with Fact Verification, called XMDFaVer, that detects online misinformation through several stages: text summarization to condense lengthy news articles into short claims, a question generation and answering model to clarify the claim with reference to the associated fact pool, and finally a classifier to assess the claim against the factual data and discern the authenticity of the news articles. XMDFaVer demonstrates strong and consistent performance across most test cases, outperforming standard and fact-based baselines. The results also show that using more fact articles improves detection accuracy, confirming the effectiveness and robustness of the approach.

1. Introduction

False information, including disinformation and misinformation, is one of society's greatest challenges. Misinformation is well known for its ability to spread quickly and widely online and its capacity to mislead and disrupt what is taken for granted in society [1]. To combat this, there are two main aspects: misinformation classification and fact-based verification. Misinformation classification aims to categorize news articles into different types of misinformation based on their characteristics, including the level of falsity. Fact-based verification is closely related to misinformation detection and is one of the most extensively studied and current topics in natural language processing [2]. This process checks the trustworthiness of claims or pieces of information and assesses the accuracy and credibility of such claims through fact-checking.

Although misinformation can be identified by the content of news only [3], external fact data can also be utilized to better detect misinformation. Additionally, one of the biggest challenges in automatic misinformation detection is the lack of annotated data for training the detection systems. Fact-checking is one of the most widely used techniques related to misinformation detection [4]. The growth of fact-checking revolves around the emergence of human-annotated data [5], the development of fact-checking websites

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¹ <https://anonymous.4open.science/r/Misinformation-detection-using-automatic-fact-based-news-verification-65D3>.

including PolitiFact,² GossipCop³ and the rise of fact verification [6]. However, previous works focus on fact-based verification/checking that judges whether a claim is true from evidence, rather than misinformation detection that classifies the misinformation from the text. This makes detecting misinformation difficult, as the data to be classified in misinformation detection are always news articles or documents, while data in fact-based verification involves short claims and related evidence. Even though misinformation detection can take advantage of fact-based verification, the nature of the data involved requires different approaches. Misinformation detection has to deal with the diversity and complexity of texts. News articles that need to be classified can be long documents that may contain multiple aspects, perspectives, and layers of information. Fact-checking, on the other hand, focuses on verifying the accuracy of a short claim or statement along with supporting evidence and requires more concise and focused analysis. These differences will require some approaches to distill long and complex news articles into concise pieces of information.

While there are potential crossovers between misinformation detection and fact-based verification, each task requires tailored methods and techniques to effectively address its unique challenges. For automatic misinformation classification tasks, the data includes news articles with labels that indicate true and fake. For automatic fact-based verification tasks, data consists of three parts: claims (pieces or summaries of the news articles), related factual information, and labels that include supported and refuted.

There are a number of automatic misinformation classification approaches that make use of machine learning [7], deep learning [8], and large language models (LLMs) [9]. While these advances have propelled the field forward, a notable gap remains when it comes to applying fact-based verification models directly to the task of detecting misinformation [10]. This gap stems from several key challenges inherent in misinformation. One of the key challenges in using fact-based verification to improve misinformation classification is the disparity in the length of input sequences between fact verification models and misinformation detection tasks. Misinformation can vary widely in content length, structure, and language style, which presents a unique challenge for traditional fact verification methods that are usually tailored for more concise and structured factual statements. Addressing this gap is crucial for effectively adapting fact-based verification techniques to the complexities of misinformation. Another significant gap is the availability and quality of training data that utilizes fact-based verification to train models that can accurately detect misinformation. A significant challenge is obtaining a comprehensive and diverse dataset that includes factual articles alongside original news content explicitly designed for training models to accurately detect misinformation.

To address these challenges, we first develop and collect two datasets. One is an extended AveriTec dataset [11] and the other is a newly collected Misbar dataset. We then propose a novel eXplainable Misinformation Detection method with Fact Verification against factual data, called XMDFaVer. This method uses text summarization to condense news articles while preserving important information and key aspects. Besides, often the facts relevant to a news article may not be fully known, which can impact the trustworthiness of the news. Therefore, we employ an information retrieval method to retrieve the most relevant fact articles to the news article from a fact pool. This is then followed by a question generation and answering component that enables automatic fact verification to support online misinformation detection. In this work, explainability refers to the ability to trace and verify the decision-making process by examining the supporting evidence, rather than focusing on interpreting the internal mechanisms of the classifiers. This allows the outcomes to be checked and validated against the retrieved and summarized factual information.

Thus, the main contributions of this work are as follows:

- A novel misinformation detection method, XMDFaVer, is proposed that combines text summarization and automatic fact-based news verification using an explainable question and answer mechanism.
- The efficacy of information retrieval-based claim verification is analyzed with reference to varying numbers of top and relevant fact articles on two curated datasets.
- A new misinformation verification and detection dataset, Misbar, is collected and released, which contains 4051 news documents¹. Besides, the Extended-AveriTec dataset is collected using references to claim and fact articles.

The rest of the article is organized as follows. Section 2 reviews the related works in the literature, including misinformation detection methods and fact-based news verification methods in natural language processing. Section 3 discusses the datasets we curated, Extended AveriTec and Misbar. In Section 4, we present the XMDFaVer method. Section 5 contains the details of the experiments, including the experimental settings and evaluation. Section 6 discusses and explains the main findings of this paper and is followed by concluding remarks in Section 7.

2. Related work

2.1. Misinformation detection

The previous studies on misinformation detection methods primarily concentrate on two aspects: content-based approaches [12], which classify false information by analyzing the news articles' contents, and context-based approaches [13], which focus on interactions in social networks between users. In this work, we employ a content-based approach to identify misinformation. A neural network model is presented that demonstrates the ability to identify true as well as false statements while also presenting self-evidence [12]. The model includes an attention mechanism that analyzes the text and highlights the more relevant words that are present in it to label outcomes. A BERT-based misinformation detection approach named BAKE is proposed to detect the stance of the body towards the

² <https://www.politifact.com/>.

³ <http://www.gossipcop.com/>.

headline [14]. The proposed method is first built with a pre-trained BERT and then fine-tuned with the Fake News Challenge stage #1 (FNC-1) dataset [15]. An improved version of BAKE, exBAKE, was introduced, which utilized external data from CNN and DailyMail for pre-training. The data are classified into four classes, including ‘agree’, ‘disagree’, ‘discuss’, and ‘unrelated’, using weighted cross-entropy as the loss function. Improvements in misinformation detection can be achieved through advanced feature engineering. Previous studies [16] have demonstrated the effectiveness of linguistic features in tackling this task. However, the success of these approaches is highly dependent on the quality of the input features extracted from the text. For example, nuanced linguistic cues and patterns must be accurately identified and exploited to improve detection accuracy. Extracting and exploiting the rich semantic, syntactic, and lexical information in the content is still a challenge in misinformation detection [17]. Meanwhile, current state-of-the-art (SOTA) methods have further advanced misinformation detection by leveraging external resources or meta-information. Techniques such as Sadhan [18], Score-Based Multi-Source Fake News Detection [19], and HetTransformer [20] incorporate additional data such as article titles, subjects, authors, and domains. These external resources are proven to provide valuable context and improve the robustness of the detection models. However, acquiring and integrating these resources can be both costly and labor-intensive.

2.2. Fact-based verification

Fact-based verification has made rapid progress with the development of pre-trained language models and the release of large-scale datasets [21]. Dong et al. [22] proposed a pre-trained language model for fact-based verification of textual statements given evidence from tables. Their proposed model is built upon the Table Parsing (TAPAS) model, incorporating efficient encoding techniques for table content, structure, and numerical comparison information to improve the performance of the model in table-based fact-based verification tasks. In the paper by Jobanputra et al. [23], an unsupervised approach based on question–answering was proposed for fact-based verification of the short claims based on the evidence. This work involves the generation of questions for a given claim, predicting the masked span, and subsequently comparing it to the ground truth answer to ascertain the label classification. Pan et al. [24] proposed Question–Answering for Claim Generation (QACG), a framework designed to train a fact-based verification model. The framework leverages automatically generated claims derived from evidence sourced from Wikipedia, which can be categorized as supported, refuted, or unverifiable. QACG generates question–answer pairs from the evidence and converts them into diverse types of claims. These claims serve as the foundation for the development of a robust fact-based verification model. While recent progress has helped move the field forward, a clear gap remains in applying fact-based verification methods directly to misinformation detection [10]. One main challenge is the difference in input length: fact verification models are usually designed for short and clear claims, while misinformation can be long, unstructured, and written in many different ways. This mismatch makes it hard to apply existing methods effectively. Closing this gap is important to better utilize fact-based verification for detecting misinformation.

2.3. Existing datasets

The existing datasets relevant to the task of misinformation classification can be categorized into two classes, one for online misinformation detection and the other for fact verification. There are several benchmark datasets for misinformation detection. FakeNewsNet [5] was initially released in 2019 and is widely used in false information detection tasks. It contains news articles from the PolitiFact⁴ and GossipCop⁵ websites. In FakeNewsNet (PolitiFact), there are 821 articles with 457 true news and 364 false news. FakeNewsNet (GossipCop) has 19,717 articles with 14,953 true news and 4764 false news. Kaggle (Jruvika) [5] and Kaggle (C) [25] are open datasets in Kaggle that are used in online misinformation detection tasks [26]. Kaggle (Jruvika) [5] contains 4 attributes, including URLs, Headline, Body, and Label of 3988 articles. Kaggle (C) [25] contains 25,954 articles with 12,726 true articles and 13,228 false articles. In terms of fact verification, FEVER [27] is one of the most popular datasets and contains 185,445 labeled claims where the facts have been extracted from Wikipedia. LIAR [21] has 12,836 short claims with their speakers, contexts/venues, labels, and justifications.

Although there are several widely used benchmark datasets, the fact-based datasets in the literature do not contain the original news articles of the claim and the datasets for online misinformation detection do not contain the fact articles [28].

3. Curated datasets

We collected and curated two datasets for our proposed XMDFaVer. The first dataset is an extended version of AveriTec [11], and the second is a newly collected Misbar dataset. Fact-justification articles are generally not readily available alongside the news articles that require verification. We take the step of isolating these justification articles from each dataset into a separate data collection of facts (fact base) for fact-based verification. We split each dataset into two collections, one is the news base, which contains the complete news articles and their labels, and the other is the fact base, which contains the entire set of fact articles. This separation allows us to better account for the inherent variations in the sourcing and presentation of this critical information during the fact-based verification process in real scenarios.

⁴ <https://www.politifact.com/>.

⁵ <http://www.gossipcop.com/>.

3.1. Extended-AveriTec

AveriTec [11] is a comprehensive resource designed for real-world claim verification using web-based evidence [11]. This dataset encompasses fact-checks conducted by 50 different organizations. Each claim within the dataset has accessible online evidence. Additionally, the dataset provides textual justifications that elucidate how the available evidence converges to form a definitive verdict. The original dataset has 4568 real-world claims. In this work, we focus on the full articles from which the claims have been sourced. Only 1750 claims in this dataset have corresponding full news articles available. We crawled the relevant links and collected the full news articles as well as the fact articles using the URLs provided. While the original AveriTec is a multi-class dataset, we adopted two settings of what we refer to as Extended-AveriTec: a multi-class Extended-AveriTec and a binary class Extended-AveriTec. In the binary dataset, the false class is contributed by 'Refuted', 'Conflicting Evidence/Cherry-picking', and 'Not Enough Evidence'. The real class contains 'Supported', while the multi-class Extended-AveriTec includes all 4 classes. In addition, it is worth noting that in real scenarios, fact-justification articles are not always readily available alongside the news articles that need to be verified. Therefore, we have taken the step of isolating these justification articles in a separate dataset of facts (fact pool) for fact-based verification. This

News

Elon Musk Is Not Launching a Stock And Bitcoin Crypto Trading Platform

Wesam Abo Marq

6th August 2023 | Last update 8:18 pm – 6th August 2023

Emerging story

Claim Article

Recently, social media users circulated a claim alleging that billionaire and business magnate Elon Musk and his platform X will launch their own stock and bitcoin crypto trading platform.

One user tweeted, "Elon Musk and his platform X to launch their own stock trading & Bitcoin Crypto trading platform. I'm so sorry for those who are not into crypto currency you are really missing a lot."

✕
Fake

Label

Misbar's Analysis

Justification Article

Misbar investigated the circulating claim and found it to be fake.

Elon Musk denied launching a stock trading and cryptocurrency trading platform.

Elon Musk Refuted the Claim

On his "X" account, **Elon Musk** stated that he has no intentions of introducing crypto tokens. This statement came in response to a post discussing dubious X and Twitter-based digital currencies.

In reply to a post by DogeDesigner on August 5, Elon Musk addressed the concern regarding scam tokens such as (X) and TwitterDAO (TWITTER) making false claims about their association with his social media platform. DogeDesigner had warned the crypto community to exercise caution with articles related to scam tokens and made it clear that neither Musk nor X had ever initiated a crypto token. Musk further asserted in his response, "And we never will."

And we never will

— Elon Musk (@elonmusk) August 5, 2023

Fig. 1. An instance of the Misbar dataset.

Table 1

Statistics of extended AveriTec. Note: R is 'Refuted', Cf is 'Conflicting Evidence/Cherry-picking', N is 'Not Enough Evidence', and Sup is 'Supported'.

Multi-class	R/Cf/N/Sup
Distribution of Multi-class	1007/104/160/479
Binary-class	False/Real
Distribution of Binary-class	1271/479
#Words in longest article	19,365
Median length	826

Table 2

Statistics of Misbar. Note: F is 'Fake', Ml is 'Misleading', Sec is 'Selective', Sus is 'Suspicious', Mt is 'Myth', Co is 'Commotion', St is 'Satire', and T is 'True'.

Multi-class	F/Ml/Sec/Sus/Mt/Co/St/T
Distribution of Multi-class	1775/1518/107/108/14/155/114/260
Binary-class	False/Real
Distribution of Binary-class	3791/260
#Words in longest article	373
Median length	148

separation allows us to better account for the inherent variations in the sourcing and presentation of this critical information during the fact-based verification process in real scenarios. We establish the exploratory analysis of the datasets, including the word count, character count, and word cloud of whole claim articles, whole fact articles, and summarized claim articles in [Appendix A](#).

3.2. Misbar

Misbar is a fact-checking platform⁶ that uses a rating system for labeling published articles on public news websites. The Misbar platform supports both English and Arabic misinformation detection tasks. This paper employs the English version of Misbar, while the platform also provides Arabic content [29]. The developers of Misbar claim that they fact-check important and newsworthy statements, emphasizing accuracy over opinions. We searched Misbar for all English news already labeled by experts from September 10, 2019, to August 18, 2023. There are 4051 news articles with justification articles in this dataset, and there are eight labels: 'True', 'Fake', 'Misleading', 'Selective', 'Suspicious', 'Myth', 'Commotion', and 'Satire'. Similar to Extended-AveriTec, we adopted two dataset settings: multi-class and binary. For binary, the false class contains 'Fake', 'Misleading', 'Selective', 'Suspicious', 'Myth', 'Commotion', and 'Satire'. [Fig. 1](#) illustrates one instance from the Misbar dataset. The 'Emerging story' contains the content of the news (claim) article, the label is given by experts, and 'Misbar's Analysis' is the justification article that supports the label given by the experts. Similar to AveriTec, we have separated and put justification articles into a fact pool for this dataset ([Table 1](#)).

A custom crawler was developed to systematically extract English-language articles labeled by experts between September 10, 2019, and August 18, 2023. This resulted in a collection of 4051 news articles paired with corresponding justification articles. The crawler parsed structured HTML and metadata to retrieve the full text of both claims and justifications, along with their publication timestamps, verdict labels, and topical categories. Articles with missing content or fewer than 100 words were excluded to ensure adequate context for downstream processing, such as summarization and question-answer generation.

All verdict labels in the Misbar dataset are sourced directly from expert annotations on the platform, which are considered reliable due to Misbar's reputation as a professional third-party fact-checking organization. As shown in [Table 2](#), the dataset includes eight fine-grained labels: True, Fake, Misleading, Selective, Suspicious, Myth, Commotion, and Satire.

Following the same structure as the Extended-AveriTec dataset, we adopted binary classification settings for evaluation. In the binary setting, the False class is formed by aggregating all non-True labels: Fake, Misleading, Selective, Suspicious, Myth, Commotion, and Satire. We establish the exploratory analysis of the datasets, including the word count, character count, and word cloud of whole claim articles, whole fact articles, and summarized claim articles in [Appendix A](#).

4. Methodology

4.1. Problem definition and model overview

Input. The inputs of the proposed architecture consist of two parts. One is composed of full news articles in the given full-news dataset $A = a_1, a_2, \dots, a_n$. The other is composed of full-fact articles $FA = fa_1, fa_2, \dots, fa_z$.

Output. The outputs of the proposed model are the predicted labels of news articles $\hat{Y} = \hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$, which can be true (1) or fake (0).

Given a training dataset of news articles with their true labels Y , the goal is to learn a model based on the given news articles A .

⁶ <https://misbar.com/en/factcheck>.

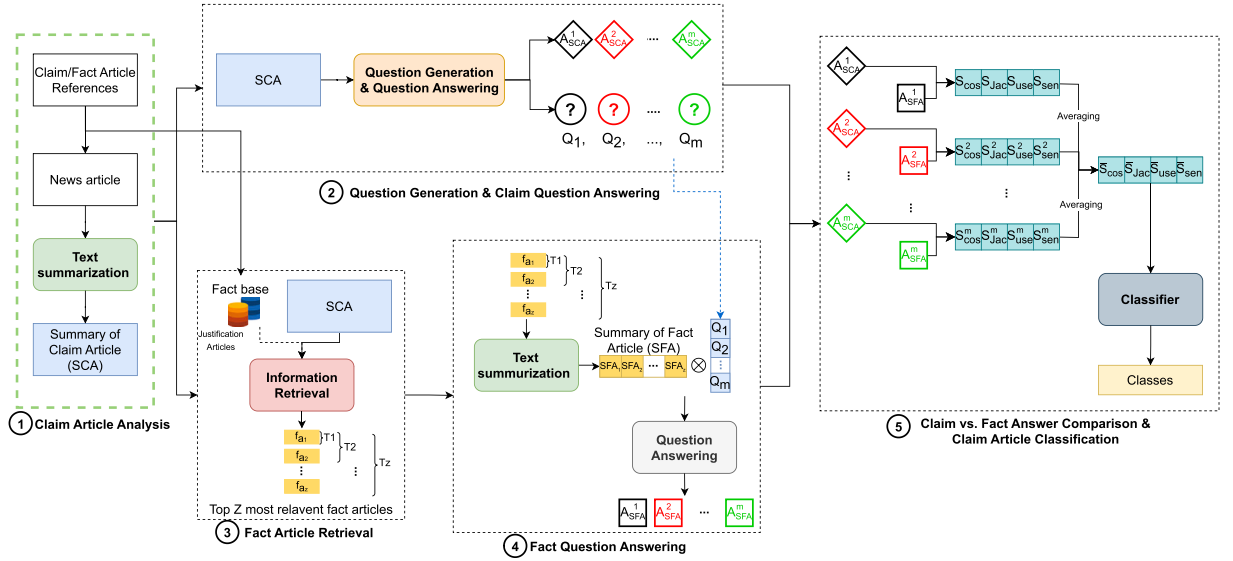


Fig. 2. The pipeline architecture of XMDFaVer starts with claim/fact articles that are first summarized. Fact document retrieval is used to find the most relevant articles from a fact pool, and then question generation and answering are utilized for factual analysis of claim articles. Finally, the new article is classified using the features extracted after the factual analysis of answers generated from the news and fact articles.

In a more general scenario, the news articles are published without corresponding facts. To address this, the approach first separates the claim articles (A) and fact articles (FA) from each dataset into a news base and a fact base, respectively, to reduce the inferences. For the extended AveriTec dataset, an additional web-crawling process is conducted to collect claim and fact articles based on the references, as the original AveriTec dataset only contains human-annotated short claims. Given the length of both claim and fact articles, they cannot be directly applied to LLMs. Therefore, claim article analysis is the first stage of the proposed architecture, following article crawling. This stage focuses on preprocessing the text to a length suitable for LLMs, enabling effective question generation and answering. An aspect-based BART model for text summarization is used to condense long news articles into short summaries consisting of a few sentences, given the considerable length of the claim and fact articles [26]. Once the claim article summaries are created, subsequent steps enable a more precise and interpretable process of utilizing factual evidence to verify the verdict of news articles. For the news base, question-answer pairs are generated for each summarized claim article. This step ensures that the generated question-answer pairs can be used to facilitate the comparison between claims and factual information. Pre-trained question-answer generation models are utilized to extract relevant questions from the news set, eliminating the need to develop such models from scratch. With the key components extracted from the news base, the process subsequently advances to the fact base. Given the assumption that there is no directly linked factual information associated with the claim articles, a document retrieval process is employed to identify the most similar fact articles from the fact base that best match the summarized claim articles. A pre-trained question-answer generator is subsequently utilized to produce answers to the same set of questions, this time using the retrieved fact articles instead of the original news base. The pre-trained question-answer generator is then applied to generate answers to the same questions from the news base, but this time using the retrieved fact articles. In the final stage, the answer pairs are evaluated, and the news articles are classified. Multiple text similarity measures are applied to compare the answers generated from the summarized news articles and those from the fact base. The resulting data is then partitioned into training and testing sets, upon which classification models are trained to predict veracity labels on the test set. A more detailed explanation of the different components of XMDFaVer in Fig. 2 is given in the following sections.

4.2. Claim article analysis

This is the first stage of XMDFaVer as shown in Fig. 2. The goal in this stage is to distill the essence of a news article, ensuring that the core message can be transferred in the following stages. An aspect-based text summarization method proposed in Ref. [26] is employed to summarize claim articles into Summary of Claim Article, $SCA = sca_1, sca_2, \dots, sca_n$. This summarization method is based on aspects, which are not known *a priori* and learned from document data. The aspects contain crucial information within the document, and in this work, their inclusion is instrumental in constructing a summary that accurately represents a news article. This condensation process ensures that the lengthy news articles are conveyed in a concise and informative manner, and it also makes it possible in the next steps to make effective use of LLMs for question and answer analysis purposes.

4.3. Question generation and claim question answering

In this step, a question generator is employed as a part of question generation and answering by using LLMs, including Llama2 and GPT3.5. These models are capable of processing and understanding vast amounts of textual information from both the fact base

and the news base. By generating relevant questions based on the content of the news articles, the models help break down claims into explainable and answerable questions that can be verified individually. This approach allows for a more focused and systematic way of checking the accuracy of the information.

The question generation and answering mechanism is particularly useful because it enables XMDFaVer to compare the news content with facts directly. The prompt used to guide question generation was manually crafted based on prior work in question–answer generation and further refined through prompt engineering [30]. The final version includes clear instructions to extract fact-checkable questions from summaries while ensuring diversity in phrasing and coverage of key claim aspects. By asking specific questions from the summarized claim articles, it can later be applied to verify the answers from the claim articles and answers retrieved from the fact base. The model can identify whether the claims made in the article align with established truths. This step enhances the verification process, ensuring that relevant information about the claim is thoroughly checked.

Additionally, using question generation and answering can enable the explainability of the process. When the system generates and answers questions, it provides a transparent method for fact-checking. The questions highlight which parts of the news are being scrutinized, and the answers offer a clear rationale for why a claim is true or false.

The prompt we used to generate the question and answer pair is,

“Generate 5 pairs of questions and no more than 5 words of answers from the given TEXT.

TEXT: {text}”

where the *TEXT* is the *SCA* from the stage 1.

With each claim summary *SCA*, the model returns a list of questions $Q = q_1, q_2, \dots, q_m$ that contains the generated questions based on the summary *SCA* and corresponding answers denoted as $A_{SCA} = a_{SCA}^1, a_{SCA}^2, \dots, a_{SCA}^m$. The generated answers A_{SCA} from the news base are then transferred to stage 6 to be verified by the factual answers from stage 5. Fig. 3 shows an example of the generated questions and corresponding answers from the news base of the extended-AveriTec dataset using LLama2. The summarized claim in this example states that “Moon and Mars will appear to be the same size in the night sky on August 27, 2020.” From this claim, the system generates five distinct questions aimed at extracting key details.

The first question (Q1) asks, “When will Moon and Mars appear the same size in the night sky?” to which the corresponding answer (a_{SCA}^1) is “August 27, 2020,” highlighting the specific date mentioned in the claim. The second question (Q2) shifts focus to the celestial bodies involved, asking, “Which celestial bodies will appear the same size on August 27, 2020?” The system correctly answers this with “Moon and Mars,” directly referencing the entities mentioned in the claim.

The third question (Q3) explores the location, inquiring, “Where will Moon and Mars appear the same size?” The answer given is “In the night sky,” which reaffirms the setting described in the summarized claim. Question four (Q4) delves into the appearance of the celestial bodies on the specific date, asking, “What will Moon and Mars look like on August 27, 2020?” to which the system answers “The same size,” restating the visual observation that was mentioned. Finally, the fifth question (Q5) revisits the timing, asking, “On which date will Moon and Mars appear to be the same size?” and once again, the system responds with “August 27, 2020,” reinforcing the key date.

This figure demonstrates the ability of the LLMs to generate contextually relevant questions based on a given claim and also to provide accurate, concise answers. The process allows for better extraction of essential information and presents the claim in a more explainable format, which is critical for tasks like misinformation detection or fact verification.

4.4. Fact article retrieval

As discussed earlier, the links between fact articles and claim articles are purposely removed in the process of data curation so that the dataset resembles the real-world scenario of misinformation detection. In this fourth stage, it is necessary to use document retrieval, seeking to procure substantiating evidence for each claim summary. This procedure commences by employing Term Frequency-Inverse Document Frequency (TF-IDF) to convert the fact articles stored in the fact base and the claim summaries into vectors. Subsequently, a cosine similarity function is utilized to determine the ranking of fact articles in order of their resemblance to each claim summary.

To find relevant fact articles in the fact base with reference to claim summaries from the news base using TF-IDF, we follow a systematic process involving preprocessing, vectorization, and similarity calculation. The process starts by tokenizing the text, splitting it into individual words, normalizing it by converting all text to lowercase, removing punctuation, and eliminating common stopwords that do not contribute significant meaning. Stemming and lemmatization are also applied to reduce the number of words in their base forms, which helps improve the accuracy of matches.

Once the data is preprocessed, the next step is to calculate the TF-IDF for the fact articles in the fact base, starting by combining all the articles into a single corpus. Term Frequency (TF) is calculated.

Each article is converted into a vector using TF-IDF. This vector representation captures the importance of each term in relation to the entire corpus of fact articles, allowing for a quantitative measure of relevance.

In the following Section 5, we delve into various scenarios, including the examination of the top 1 and top 2 most relevant fact articles. We also consider the temporal leaks (time lag) in the data [11]. Specifically, a claim article published on a given day—e.g., 15 January 2024—should not be verified using any fact article published after that date. To address this, we apply time lag filtering at the daily level by restricting the Fact Article Retrieval step to include only fact articles published strictly prior to the claim article’s publication date.

Summarized Claim	Moon and Mars will appear to be the same size in the night sky on August 27, 2020.
Generated Questions	Corresponding Answers
Q1 When will Moon and Mars appear the same size in the night sky?	A_{SCA}^1 August 27, 2020.
Q2 What celestial bodies will appear the same size on August 27, 2020?	A_{SCA}^2 Moon and Mars.
Q3 Where will Moon and Mars appear the same size?	A_{SCA}^3 In the night sky.
Q4 What will Moon and Mars look like on August 27, 2020?	A_{SCA}^4 The same size.
Q5 On which date will Moon and Mars appear to be the same size?	A_{SCA}^5 August 27, 2020.

Fig. 3. An example of the generated questions and corresponding answers from the extended-AveriTec dataset using LLama2.

4.5. Fact question answering

As shown in Step 4 of Fig. 2, the same LLMs utilized in the Claim Question–Answering stage are leveraged. Given that fact articles can be extensive and lengthy documents, we first employ an aspect-based text summarization method to distill these fact articles into a summary of the fact article, Summary of Fact Article, denoted as $SFA = sfa_1, sfa_2, \dots, sfa_z$. This condensation process ensures that the evidence is conveyed in a concise and informative manner, and also makes the LLM prompt’s size acceptable and fair over the algorithms, enhancing the efficiency of the fact-checking endeavor.

In the subsequent phase of this step, the same LLM utilized in the Claim Question–Answering stage is leveraged. In this step, the model is prompted with a combination of generated questions based on each summary SFA . The LLM then processes this input, utilizing its expansive knowledge and language capabilities, to produce a set of corresponding fact-based answers for the posed questions.⁷ These answers are assembled into a collection designated as $A_{SFA} = a_{sfa}^1, a_{sfa}^2, \dots, a_{sfa}^m$, effectively serving as the output of this pivotal stage.

The prompt we use is,

“Answer the Question in no more than 5 words by looking into the Facts.

Question: {question}

Facts: {facts}”

Fig. 4 illustrates an example of the generated answers from the fact base. It contains generated questions based on an SCA with their corresponding answers from both the news base and the fact base. The SCA states: “Moon and Mars will appear to be the same size in the night sky on August 27, 2020.”

In the Generated Questions column (highlighted in orange), five questions are presented, each asking differently about the SCA. These include when and where the Moon and Mars will appear to be the same size, which celestial bodies will be involved, and what they will look like on the mentioned date. The questions aim to probe the specific details of the claim to assess its accuracy.

The second column, labeled Corresponding Answers (Claim Article) (shown in purple), contains answers derived directly from the SCA. These answers reiterate the original information provided in the claim, stating that on August 27, 2020, both the Moon and Mars will appear to be the same size. For instance, A_{SCA}^1 indicates that the event will occur on August 27, while A_{SCA}^2 states that both celestial bodies involved will be the Moon and Mars.

In contrast, the Corresponding Answers (Fact Article) column (in gray) provides fact-checked answers that often refute the original claim. Each fact-based answer clarifies that the Moon and Mars do not, and will not, appear the same size in the night sky. For example, A_{SFA}^1 explicitly states that the Moon and Mars do not appear the same size, while A_{SFA}^2 points out that there are no celestial bodies that will appear the same size on August 27, 2020. These fact-based answers highlight the discrepancies between the claim and actual scientific evidence.

4.6. Answer analysis & claim article classification

Once the answers from both the news base and fact base are generated based on the questions, it leads to the final stage of verification. In this stage, there are two steps: answer checking and prediction. Answer checking is the first step, focusing on evaluating the quality and accuracy of answers derived from the text summaries and corresponding justifications. Once the answers have been checked and validated, the next step is prediction. This step involves making a final judgment or classification of the news article based on the validated information.

⁷ Manual checking ensures no empty outcomes.

Summarized Claim	Moon and Mars will appear to be the same size in the night sky on August 27, 2020.		
Generated Questions	Corresponding Answers	Fact Answers	
Q1 When will Moon and Mars appear the same size in the night sky?	A_{SCA}^1 August 27, 2020.	A_{SFA}^1 Moon and Mars do not appear the same size in the night sky.	
Q2 What celestial bodies will appear the same size on August 27, 2020?	A_{SCA}^2 Moon and Mars.	A_{SFA}^2 Based on the given facts, there are no celestial bodies that will appear the same size on August 27, 2020.	
Q3 Where will Moon and Mars appear the same size?	A_{SCA}^3 In the night sky.	A_{SFA}^3 The Moon and Mars do not appear the same size in any situation.	
Q4 What will Moon and Mars look like on August 27, 2020?	A_{SCA}^4 The same size.	A_{SFA}^4 Based on the given facts, it is not possible to determine what the Moon and Mars will look like on August 27.	
Q5 On which date will Moon and Mars appear to be the same size?	A_{SCA}^5 August 27, 2020.	A_{SFA}^4 The given facts do not provide any information about when the Moon and Mars will appear the same size in the night sky.	

Fig. 4. An example of the prompt to generate answers from the fact base using Llama-2.

Answer checking. is applied to evaluate the answers from summaries and justifications. In this step, each generated answer from the news base A_{SCA} is cross-referenced with the answer generated from the fact base A_{SFA} , checking whether A_{SCA} aligns with the A_{SFA} and supports fact validation processes. This step is crucial in assessing the consistency between A_{SCA} and A_{SFA} , evaluating the overlap and similarity between the A_{SCA} and A_{SFA} . We consider four evaluation methods, cosine similarity [31], Jaccard similarity [32,33], Universal Sentence Encoder semantic similarity [34], and semantic similarity from [35].

- **Cosine Similarity:** It measures the cosine of the angle between two non-zero vectors, providing insight into their similarity. In this stage, cosine similarity helps determine how closely the claim answer A_{SCA} aligns with the fact answer A_{SFA} by comparing the vectors of words or tokens. A value of 1 means the vectors are identical, while a value of 0 means they are orthogonal, or completely dissimilar. Cosine similarity can compare the textual structure between the answers, ensuring that they are similar in content even if the exact wording differs.
- **Jaccard Similarity:** This method evaluates the similarity between two sets by comparing their intersections and unions. In this stage, Jaccard similarity assesses the overlap of words between two answers. The Jaccard index is calculated by dividing the size of the intersection of the two sets by the size of their union. A high Jaccard similarity indicates that the answers share a large number of common words or phrases. This method is effective in checking for the presence of key terms or phrases that should appear in both the claim answers and the fact answers, focusing on overlap in vocabulary.
- **Universal Sentence Encoder Semantic Similarity:** Leveraging Google's Universal Sentence Encoder, this approach gauges the semantic similarity between sentences. In this stage, this method focuses on the overall meaning of the sentences rather than just word-level similarities. It converts sentences into high-dimensional embeddings, capturing their semantic meaning. By comparing the embeddings of the claim answers and the fact answers, it evaluates whether the two sentences have similar meanings, even if the wording is different. This approach allows for a more nuanced comparison, accounting for synonyms, paraphrases, and broader linguistic similarities that may not be captured by methods like cosine or Jaccard similarity.
- **Semantic Similarity from [35]:** A semantic similarity approach relies on the concept that a sentence's meaning is shaped by its surrounding context. It quantifies sentence similarity by assessing the likelihood of generating two sentences within a shared context. The authors use attention mechanisms to weigh the importance of different sentences. The model is trained on pairs of sentences, where the task is to predict their similarity score based on both the sentence content and its context. The paper evaluates multiple training setups, including fine-tuning large pre-trained language models to generate contextualized sentence embeddings that reflect similarity in meaning more accurately.

These four evaluation methods calculate the similarity scores of the answers from summaries and the answers from justifications. As the number of questions for each summary can vary, we average the similarity scores of answers for each similarity measure to determine the score of the summary based on Eq. (1) as follows,

$$\bar{S} = \frac{1}{m} \sum_{i=1}^m S_{eval_method}^i, \quad (1)$$

where $\bar{S}$ is the averaged similarity score, S^i is each similarity score of answers, and m is the number of the question–answer pairs for each claim. There is $N \times \bar{S}$, where N is the number of similarity measures.

Prediction. After completing the answer-checking phase, the next critical step is the prediction phase, where we aim to determine the final classification of a claim based on the validated answers. The main goal in this stage is to leverage machine learning models to predict whether a claim is true, false, or misleading. For this purpose, the input data consists of both the bag-of-words vectors and

the similarity scores derived from various similarity measures. These features are essential for building the models and guiding the prediction process.

In order to train and evaluate the machine learning models effectively, the dataset is split into two subsets: 80 % of the data is used for training, while the remaining 20 % is reserved for testing. This 80/20 split ensures that the models are trained on a significant portion of the data while being evaluated on unseen examples to assess their performance and generalization capabilities.

In this prediction phase, we utilize four state-of-the-art classification models: Logistic Regression (LR), Support Vector Machine (SVM), Gradient Boosting (GB), and Random Forest (RF). These models are chosen for their proven effectiveness in both binary and multi-class classification tasks. As there are two versions of the AveriTec dataset, one for binary classification and the other for multi-class classification, the models are applied accordingly to handle both types of classification problems.

For binary classification, LR is employed using the Sigmoid function to model the probability of a claim being true or false. The Sigmoid function is ideal for binary classification as it maps the predicted output to a probability between 0 and 1. In the case of SVM, it finds the optimal hyperplane to separate data points from the other class. GB for binary classification optimizes the model using binary cross-entropy as the loss function. For RF, it predicts the majority vote across trees.

For multi-class classification, LR is extended using the Softmax function. This approach, known as multinomial logistic regression, allows the model to predict the probabilities of multiple classes, assigning a claim to the class with the highest probability. Softmax is crucial when handling classification tasks with more than two possible outcomes. In the case of a multi-class SVM, the model uses the One-vs-Many strategy. This involves constructing a binary classifier for every possible pair of classes. When making predictions, each classifier votes for one of the two classes it was trained to distinguish, and the final classification is determined based on the class that receives the majority of votes or has the highest score. GB for multi-class classification uses cross-entropy as the loss function. The model constructs separate decision trees for each class, with each tree designed to predict the probability that a given claim belongs to that specific class. The class with the highest accumulated score across the trees is selected as the final prediction. Finally, for multi-class Random Forest RF, the algorithm works by building multiple decision trees. Each tree predicts the class of a claim, and the final classification is determined by the class that is most frequently predicted by the ensemble of trees. This method, often referred to as majority voting, ensures that the model's decision is based on the consensus of multiple trees, enhancing its robustness and accuracy.

Overall, these models are trained to classify claims based on various features and similarity measures, providing a comprehensive and flexible approach to both binary and multi-class classification in the context of misinformation detection.

5. Experiments

5.1. Baselines

Our evaluation framework encompasses two distinct baseline settings designed to assess the efficacy of XMDFaVer. The first setting is a standard baseline geared towards pure content-based misinformation detection, where only the textual content of news articles is utilized as input data. Within this setting, we employ supervised misinformation detection models that undergo training using labeled news articles.

In the second setting, the baseline with fact-checking configuration, we introduce justification or fact articles as supplementary input to enhance the misinformation detection model's performance. In this scenario, the input vector is constructed through a combination of three components: a) concatenation of representations where we fuse the representations of news articles and justification articles by concatenating them together, b) dot-product of representations in which we calculate the dot-product of the representations of news articles and justification articles, and c) subtraction of representations where we derive a new representation by subtracting the representations of justification articles from those of the news articles.

To encode the news article summaries into fixed-length vectors with 300 dimensions for the above representations, we employ the Doc2Vec method, a pre-trained paragraph embedding technique developed by Ref. [36]. This transformation allows us to maintain essential information while reducing dimensionality. Subsequently, we leverage well-established machine learning models known for their state-of-the-art performance in misinformation detection using fact verification [10]. These models include SVM, Logistic Regression, Gradient Boosting, and Random Forest, which serve as our classification models. These models undergo training and testing to be evaluated for their effectiveness in detecting misinformation, enhancing our understanding of their performance in both pure misinformation detection and fact-based verification scenarios. We also use the GPT3.5 pre-trained language model only as a performance reference point rather than a baseline. This is mainly due to the fact that truncating long news articles (as an input requirement for LLMs) would significantly affect the performance of such models [10], and in addition, the performance of this model tends to be consistent regardless of whether there are additional fact articles.

Another baseline classifier is a statistical classifier. In this case, we get the median, 3rd quantile, and mean values of each type of similarity scores from the train set. Then, the median, 3rd quantile, and mean values are treated as the thresholds, respectively, to discover the scores in the test set: if there are less than 50 % of the scores greater than the threshold, then return 'False' as the predicted label. Otherwise, return 'Real' as the predicted label. For example, if the value of the threshold is 0.45, and in the test set, the values of Cosine, Jaccard, Universal Sentence Encoder Semantic, and Semantic Similarity are 0.2, 0.5, 0.45, and 0.1, respectively, the classification stage will return 'False' as the predicted label since half of the scores are below the threshold of 0.45.

Table 3

Fact retrieval performance on Extended-AveriTec and Misbar datasets. Note: Accuracy@T1 and Accuracy@T2 are the accuracies of successfully finding the corresponding fact articles in the top 1 and top 2 fact articles, and MRR@1 and MRR@2 are the mean reciprocal ranks (MRR) at the top 1 and top 2 retrieved articles.

Dataset	Accuracy@T1	Accuracy@T2	MRR@1	MRR@2
Extended-AveriTec	91.07 %	98.10 %	0.9107	0.9459
Misbar	82.29 %	93.78 %	0.8229	0.8804

Table 4

Performance analysis of the proposed methods and supervised ML models on the binary-class Extended-AveriTec dataset. Note: QG, QA, FR, and ML denote question generation, question-answering, fact retrieval, and machine learning models for classification. T1 and T2 denote the summarized top 1 and top 2 fact articles, respectively. Precision, recall, and F1 are macro average precision, macro average recall, and macro average F1.

Setting	Model	Accuracy	Precision	Recall	F1
-QG/QA-FR	SVM	0.6351	0.6944	0.6351	0.6541
	LR	0.6048	0.6540	0.6048	0.6231
	GB	0.7057	0.7115	0.7057	0.7084
	RF	0.6573	0.6876	0.6573	0.6692
	GPT3.5*	0.8100	0.7533	0.7410	0.7471
-QG/QA+FR	SVM-T1	0.6171	0.6070	0.5868	0.5801
	LR-T1	0.6857	0.6217	0.6144	0.6173
	GB-T1	0.7657	0.7353	0.7128	0.7211
	RF-T1	0.7571	0.7294	0.7047	0.7132
	GPT3.5-T1*	0.8100	0.7533	0.7410	0.7471
	SVM-T2	0.5886	0.5548	0.5454	0.5397
	LR-T2	0.7589	0.7325	0.7070	0.7157
	GB-T2	0.7491	0.7187	0.6955	0.7034
	RF-T2	0.7423	0.7140	0.6894	0.6972
	GPT3.5-T2*	0.8100	0.7533	0.7410	0.7471
	XMDFaVer (Statistic-Median)-T1	0.5123	0.4932	0.4815	0.4692
	XMDFaVer (Statistic-3rd quantile)-T1	0.6007	0.4887	0.4871	0.4869
+QG/QA+FR-ML	XMDFaVer (Statistic-Mean)-T1	0.6756	0.4842	0.4939	0.4662
	XMDFaVer (Statistic-Median)-T2	0.5438	0.5494	0.5638	0.5192
	XMDFaVer (Statistic-3rd quantile)-T2	0.6154	0.5649	0.5796	0.5603
	XMDFaVer (Statistic-Mean)-T2	0.7061	0.5885	0.5663	0.5697
+QG/QA+FR+ML	XMDFaVer (Llama2-SVM)-T1	0.7240	0.6806	0.6643	0.6702
	XMDFaVer (Llama2-LR)-T1	0.7926	0.7408	0.7392	0.7400
	XMDFaVer (Llama2-GB)-T1	0.8006	0.7463	0.7492	0.7477
	XMDFaVer (Llama2-RF)-T1	0.7714	0.7262	0.7151	0.7200
	XMDFaVer (Llama2-SVM)-T2	0.7303	0.6979	0.6756	0.6827
	XMDFaVer (Llama2-LR)-T2	0.8017	0.7484	0.7507	0.7495
	XMDFaVer (Llama2-GB)-T2	0.8051	0.7546	0.7550	0.7548
	XMDFaVer (Llama2-RF)-T2	0.7669	0.7283	0.7116	0.7184
	XMDFaVer (GPT3.5-SVM)-T1	0.7886	0.7901	0.7468	0.7528
	XMDFaVer (GPT3.5-LR)-T1	0.8034	0.7534	0.7528	0.7531
	XMDFaVer (GPT3.5-GB)-T1	0.8263	0.7718	0.7833	0.7771
	XMDFaVer (GPT3.5-RF)-T1	0.8006	0.7541	0.7493	0.7516
	XMDFaVer (GPT3.5-SVM)-T2	0.8240	0.7657	0.7810	0.7726
	XMDFaVer (GPT3.5-LR)-T2	0.8257	0.7805	0.8359	0.7963
	XMDFaVer (GPT3.5-GB)-T2	0.8537	0.7861	0.8302	0.8036
	XMDFaVer (GPT3.5-RF)-T2	0.8091	0.7554	0.7603	0.7578

5.2. Evaluation

Fact retrieval. In the experiments, we consider two settings for fact document retrieval, one with only the top 1 relevant article (T1) and the other with the top 2 relevant articles (T2) due to the limit of input length for LLMs. Table 3 illustrates the performance of fact document retrieval. From the results of the Extended-AveriTec dataset, the corresponding fact articles can be successfully found in the top 2 ranks, which is 95.94 %. The missing rate is about 4 %, which shows that by selecting the top 2 fact articles, most of the corresponding facts can be retrieved successfully. For the Misbar dataset, around 90 % of the corresponding fact articles were successfully retrieved in the top 2 ranks. The mean reciprocal rank (MRR) at the top 2 retrieved articles (MRR@2) is 0.9026 and 0.8499 for Extended-AveriTec and Misbar, respectively.

Extended-AveriTec. Table 4 illustrates the results on the Extended-AveriTec dataset in the case of binary classification. The number of questions for each claim c_n was set to 5. The first part of the table presents the outcomes for various baseline models, encompassing

Table 5

Performance analysis of the proposed methods and supervised ML models on the multi-class Extended-AveriTec dataset. Note: QG, QA, FR, and ML denote question generation, question-answering, fact retrieval, and machine learning models for classification. T1 and T2 denote the summarized top 1 and top 2 fact articles, respectively. Precision, recall, and F1 are macro average precision, macro average recall, and macro average F1 on the four classes: 'Supported', 'Refuted', 'Conflicting Evidence/Cherrypicking', and 'Not Enough Evidence'. * GPT3.5 has the similar performance as GPT3.5 T1 and GPT3.5 T2 because of the input limits that truncate and fade out T1 and T2.

Setting	Model	Accuracy	Precision	Recall	F1
-QG/QA-FR	SVM	0.5057	0.1421	0.2500	0.1812
	LR	0.5097	0.1463	0.2515	0.1943
	GB	0.5086	0.1443	0.2500	0.1821
	RF	0.5058	0.1425	0.2500	0.1819
	GPT3.5*	0.5282	0.2714	0.2815	0.2763
-QG/QA+FR	SVM-T1	0.5120	0.2977	0.3518	0.3128
	LR-T1	0.5177	0.3002	0.3534	0.3156
	GB-T1	0.5166	0.2997	0.3530	0.3151
	RF-T1	0.5137	0.2971	0.3517	0.3133
	GPT3.5*	0.5282	0.2714	0.2815	0.2763
	SVM-T2	0.5127	0.3005	0.3540	0.3172
	LR-T2	0.5246	0.3018	0.3549	0.3186
	GB-T2	0.5234	0.3013	0.3545	0.3180
	RF-T2	0.5223	0.3008	0.3542	0.3175
	GPT3.5*	0.5320	0.3050	0.3571	0.3222
+QG/QA+FR+ML	XMDFaVer (Llama2-SVM)-T1	0.5349	0.3063	0.3580	0.3236
	XMDFaVer (Llama2-LR)-T1	0.5377	0.3075	0.3589	0.3250
	XMDFaVer (Llama2-GB)-T1	0.5389	0.3083	0.3594	0.3257
	XMDFaVer (Llama2-RF)-T1	0.5400	0.3088	0.3597	0.3262
	XMDFaVer (Llama2-SVM)-T2	0.5427	0.3219	0.2993	0.2980
	XMDFaVer (Llama2-LR)-T2	0.5549	0.3152	0.3648	0.3335
	XMDFaVer (Llama2-GB)-T2	0.5497	0.3130	0.3630	0.3310
	XMDFaVer (Llama2-RF)-T2	0.5469	0.3117	0.3620	0.3296
	XMDFaVer (GPT3.5-SVM)-T1	0.5486	0.3125	0.3626	0.3304
	XMDFaVer (GPT3.5-LR)-T1	0.5897	0.3304	0.3787	0.3507
	XMDFaVer (GPT3.5-GB)-T1	0.5840	0.3279	0.3762	0.3479
	XMDFaVer (GPT3.5-RF)-T1	0.5726	0.3229	0.3714	0.3422
	XMDFaVer (GPT3.5-SVM)-T2	0.5611	0.3179	0.3670	0.3366
	XMDFaVer (GPT3.5-LR)-T2	0.6964	0.4569	0.4594	0.4392
	XMDFaVer (GPT3.5-GB)-T2	0.6834	0.3913	0.4165	0.3925
	XMDFaVer (GPT3.5-RF)-T2	0.6377	0.3715	0.4001	0.3703

Table 6

F1 scores of the best XMDFaVer method and SOTA model on the AveriTec dataset. Note: S is 'Supported', R is 'Refuted', C is 'Conflicting Evidence/Cherrypicking', and N is 'Not Enough Evidence'. *XMDFaVer (GPT3.5-LR)-T2 model is applied on the same dataset as the SOTA model, which takes the existing claims and corresponding short facts in the dataset as input. F1 is a macro average measure on the Misbar dataset.

Model	S	R	C	N	F1
AveriTec [11]	0.41	0.69	0.10	0.16	0.23
XMDFaVer (GPT3.5-LR)-T2*	0.63	0.78	0.12	0.23	0.44

standard baselines and the baseline models with T1 and T2. The second part of the table shows the results of using Llama2 as the LLM for QG/QA. In the case of XMDFaVer methods, there are two different sets with varying quantities of top articles. The table illustrates that XMDFaVer methods outperform the baseline models as well as the pure GPT3.5 model. Remarkably, the proposed method employing GB with the top 2 articles stands out, achieving the highest performance. It attains 0.8537, 0.7861, 0.8302, and 0.8036 on the accuracy, macro precision, macro recall, and macro F1 score, respectively.

Table 5 illustrates the results on the Extended-AveriTec dataset in multi-class classification. Our XMDFaVer method, which employs LR with the top 2 articles, achieves the highest scores across various metrics. Specifically, XMDFaVer achieves an accuracy of 0.6960, a macro precision of 0.4569, a macro recall of 0.4594, and a macro F1 score of 0.4392. It is worth noting that adopting the top 2 fact articles can lead to improved performance compared with only the top 1 fact article in the detection of fake news.

Table 6 shows the macro F1 scores achieved by our top-performing model, GPT3.5-GB-T2, alongside the pipeline model introduced by AveriTec [11] on the multi-class AveriTec dataset. The macro F1 score of our XMDFaVer model is more than twice that of the SOTA model in this case.

Table 7

Performance analysis of the proposed methods and supervised ML models on the binary-class Misbar dataset. Note: QG, QA, FR, and ML denote question generation, question-answering, fact retrieval, and machine learning models for classification. T1 and T2 denote the summarized top 1 and top 2 fact articles, respectively. Precision, recall, and F1 are macro average precision, macro average recall, and macro average F1 on the two classes ‘true’ and ‘fake’. * GPT3.5 has the similar performance as GPT3.5 T1 and GPT3.5 T2 because of the input limits that truncate and fade out T1 and T2.

Setting	Model	Accuracy	Precision	Recall	F1
-QG/QA-FR	SVM	0.7764	0.7731	0.7694	0.7709
	LR	0.8253	0.8232	0.8192	0.8209
	GB	0.8393	0.8351	0.8338	0.8344
	RF	0.8362	0.8324	0.8305	0.8314
	GPT3.5*	0.8904	0.8752	0.7597	0.8150
-QG/QA+FR	SVM-T1	0.8281	0.8238	0.8219	0.8195
	LR-T1	0.8405	0.8397	0.8357	0.8365
	GB-T1	0.8492	0.8425	0.8418	0.8439
	RF-T1	0.8439	0.8419	0.8382	0.8398
	GPT3.5-T1*	0.8964	0.8774	0.7697	0.8200
	SVM-T2	0.8413	0.8394	0.8352	0.8377
	LR-T2	0.8468	0.8512	0.8483	0.8390
	GB-T2	0.8568	0.8469	0.8481	0.8479
	RF-T2	0.8497	0.8468	0.8401	0.8436
	GPT3.5-T2*	0.8964	0.8824	0.7907	0.8304
-QG/QA +FR	XMDFaVer (Statistic-Median)-T1	0.6782	0.5464	0.5583	0.5452
	XMDFaVer (Statistic-3rd quantile)-T1	0.7235	0.6301	0.6437	0.6351
	XMDFaVer (Statistic-Mean)-T1	0.7360	0.6510	0.6608	0.6518
	XMDFaVer (Statistic-Median)-T2	0.7617	0.6004	0.5914	0.5603
	XMDFaVer (Statistic-3rd quantile)-T2	0.8395	0.6637	0.5701	0.5790
-QG/QA+FR	XMDFaVer (Statistic-Mean)-T2	0.8622	0.6718	0.5965	0.6028
	XMDFaVer (Llama2-SVM)-T1	0.7815	0.5065	0.5213	0.4907
	XMDFaVer (Llama2-LR)-T1	0.8763	0.8778	0.8784	0.8698
	XMDFaVer (Llama2-GB)-T1	0.8884	0.8962	0.8996	0.8813
	XMDFaVer (Llama2-RF)-T1	0.9019	0.9146	0.9005	0.8994
	XMDFaVer (Llama2-SVM)-T2	0.8663	0.8631	0.8628	0.8626
	XMDFaVer (Llama2-LR)-T2	0.8840	0.8792	0.8719	0.8764
	XMDFaVer (Llama2-GB)-T2	0.9023	0.8918	0.8993	0.8982
	XMDFaVer (Llama2-RF)-T2	0.9193	0.9061	0.9057	0.9049
	XMDFaVer (GPT3.5-SVM)-T1	0.8853	0.8821	0.8763	0.8794
	XMDFaVer (GPT3.5-LR)-T1	0.9113	0.9097	0.9061	0.9029
	XMDFaVer (GPT3.5-GB)-T1	0.9224	0.9276	0.9261	0.9235
	XMDFaVer (GPT3.5-RF)-T1	0.9176	0.9103	0.9151	0.9147
	XMDFaVer (GPT3.5-SVM)-T2	0.8902	0.8941	0.8815	0.8864
	XMDFaVer (GPT3.5-LR)-T2	0.9276	0.9229	0.9177	0.9187
	XMDFaVer (GPT3.5-GB)-T2	0.9334	0.9304	0.9292	0.9298
	XMDFaVer (GPT3.5-RF)-T2	0.9259	0.9257	0.9262	0.9249

Misbar. The results for the Misbar dataset are presented in Table 7. Each claim c_n is associated with 5 questions. The table is divided into three parts to showcase different views of the evaluation. The first part displays outcomes for various baseline models, including standard baselines, as well as baseline models utilizing the top 1 and top 2 articles. The middle section introduces XMDFaVer models with a statistical classifier. These models are evaluated under the same settings as with QG/QA and FR. XMDFaVer (Statistic-Mean) achieves an accuracy of 0.8622 and an F1-score of 0.6028. The lower portion of the table demonstrates the performance of the XMDFaVer architecture, consisting of two LLMs, including Llama2 and GPT3.5. This section highlights that by using Llama2 as the LLM for QG/QA, the proposed methods consistently outperform the baseline models. Particularly noteworthy is the performance of the method employing Random Forest with the top 2 articles, achieving remarkable scores of 0.9193, 0.9061, 0.9057, and 0.9049 for accuracy, macro precision, macro recall, and macro F1 score, respectively. For XMDFaVer, using GPT3.5 as the LLM for QG/QA, the proposed method, utilizing Gradient Boosting with the top 2 articles stands out, demonstrating exceptional performance. It achieves the highest scores across various metrics, including an accuracy of 0.9334, macro precision of 0.9304, macro recall of 0.9292, and a macro F1 score of 0.9298.

Table 8 illustrates the results on the Misbar dataset in multi-class classification. Our XMDFaVer method, which employs LR with the top 2 articles, achieves the highest scores across various metrics. Specifically, XMDFaVer achieves an accuracy of 0.4921, a macro recall of 0.2017, a macro recall of 0.1529, and a macro F1 score of 0.1503.

Table 8

Performance analysis of XMDFaVer methods and supervised ML models on the multi-class Misbar dataset. Note: T1 and T2 denote the summarized top 1 and top 2 fact articles, respectively. Precision, recall, and F1 are macro average measures on the ‘True’, ‘Fake’, ‘Misleading’, ‘Selective’, ‘Suspicious’, ‘Myth’, ‘Commotion’, and ‘Satire’. * GPT3.5 has the similar performance as GPT3.5 T1 and GPT3.5 T2 because of the input limits that truncate and fade out T1 and T2.

Setting	Model	Accuracy	Precision	Recall	F1
-QG/QA-FR	SVM	0.3679	0.0915	0.1110	0.0984
	LR	0.3624	0.0894	0.1099	0.0864
	GB	0.3519	0.0876	0.1063	0.0947
	RF	0.3543	0.0882	0.1071	0.0951
	GPT3.5*	0.4407	0.1110	0.1297	0.0988
+QG/QA+FR+ML	SVM-T1	0.3996	0.1041	0.1229	0.1096
	LR-T1	0.3959	0.0992	0.1195	0.1065
	GB-T1	0.3803	0.0975	0.1189	0.1031
	RF-T1	0.3794	0.0974	0.1168	0.1016
	GPT3.5*	0.4468	0.1184	0.1325	0.1012
	SVM-T2	0.4128	0.1358	0.1351	0.1204
	LR-T2	0.4097	0.1035	0.1261	0.1192
	GB-T2	0.3991	0.1084	0.1229	0.1077
	RF-T2	0.3946	0.0996	0.1189	0.1061
	GPT3.5*	0.4456	0.1133	0.1317	0.1026
+QG/QA +FR+ML	XMDFaVer (Llama2-SVM)-T1	0.4048	0.1102	0.1228	0.1167
	XMDFaVer (Llama2-LR)-T1	0.3992	0.1108	0.1243	0.1169
	XMDFaVer (Llama2-LR)-T1	0.3992	0.1108	0.1243	0.1169
	XMDFaVer (Llama2-GB)-T1	0.3859	0.1096	0.1203	0.1127
	XMDFaVer (Llama2-RF)-T1	0.3892	0.1194	0.1256	0.1173
	XMDFaVer (Llama2-SVM)-T2	0.4359	0.1317	0.1390	0.1253
	XMDFaVer (Llama2-LR)-T2	0.4229	0.1195	0.1306	0.1270
	XMDFaVer (Llama2-GB)-T2	0.4279	0.1302	0.1316	0.1284
	XMDFaVer (Llama2-RF)-T2	0.4059	0.1301	0.1294	0.1202
	XMDFaVer (GPT3.5-SVM)-T1	0.4419	0.0821	0.1429	0.0958
	XMDFaVer (GPT3.5-LR)-T1	0.4492	0.0612	0.1429	0.0927
	XMDFaVer (GPT3.5-GB)-T1	0.4202	0.1320	0.1382	0.0921
	XMDFaVer (GPT3.5-RF)-T1	0.3920	0.1102	0.1422	0.1204
	XMDFaVer (GPT3.5-SVM)-T2	0.4712	0.1290	0.1772	0.1429
	XMDFaVer (GPT3.5-LR)-T2	0.4921	0.2017	0.1529	0.1503
	XMDFaVer (GPT3.5-GB)-T2	0.4582	0.1720	0.1724	0.1358
	XMDFaVer (GPT3.5-RF)-T2	0.4281	0.1629	0.1538	0.1388

6. Discussion

This study aims to explore the role of facts and utilize fact-based verification in online misinformation detection, including how facts and different levels of facts affect the performance of online misinformation detection. The results show the effectiveness of the proposed XMDFaVer from different perspectives.

First, the contribution of text summarization in XMDFaVer has been demonstrated to provide significant benefits. Unlike existing fact-based verification methods, which predominantly rely on short claims that require manual generation by humans, XMDFaVer leverages summarization to automatically distill lengthy news articles into concise and meaningful summaries. This automated approach reduces the reliance on manual effort while retaining the critical information necessary for effective verification, streamlining the fact-checking process. While it is true that summarization introduces some degree of information loss, it nonetheless results in overall improvements in the processing pipeline. As an abstractive approach, summarization systematically and semantically condenses the input text, aiming to preserve essential information and meaning. Despite its inherent limitations, summarization is indispensable in handling lengthy documents. Without it, truncation—an alternative often necessitated by input length constraints—proves to be significantly less effective, as it disproportionately emphasizes the beginning of the text and risks excluding critical content. Thus, summarization is a mandatory step to ensure a more balanced and comprehensive representation of the input. In this work, text summarization provides an automatic approach to combine fact-based verification of short claims with online misinformation detection.

Second, the experimental results have shown that facts can improve online misinformation detection because they provide the ground truth against the given news. Moreover, compared to detecting misinformation directly from long news articles, our proposed XMDFaVer methods exhibit superiority over baseline models in most of the test cases. While our experiments show the utility and performance-boosting capabilities of XMDFaVer when using reference fact article retrieval, we have also demonstrated that retrieving the top 2 reference fact articles is an optimal option for bringing in extra reference information for fact-based verification of news articles. To be more precise, our experiments have shown that it is possible to successfully retrieve most of the corresponding fact articles by selecting only the top 2 fact articles, which, puts together with the input, limit the most current models, remove the

Claim article	Recently, social media users circulated a claim alleging that billionaire and business magnate Elon Musk and his platform X will launch their own stock and bitcoin crypto trading platform. One user tweeted, "Elon Musk and his platform X to launch their own stock trading & Bitcoin Crypto trading platform. I'm so sorry for those who are not into crypto currency you are really missing a lot."	
Summarized claim	Claim circulating on social media alleges Elon Musk and his platform X will launch a stock and bitcoin crypto trading platform. Users express anticipation, suggesting potential opportunities for those involved in cryptocurrency.	
Justification article	Elon Musk denied launching a stock trading and cryptocurrency trading platform. Elon Musk Refuted the Claim On his "X" account, Elon Musk stated that he has no intentions of introducing crypto tokens. This statement came in response to a post discussing dubious X and Twitter-based digital currencies. ...	
Summarized justification	Elon Musk denied launching a stock and cryptocurrency trading platform, refuting claims on Twitter. He clarified that no work is being done on such a platform and emphasized that neither he nor his platform X has initiated any crypto tokens. Despite Musk's denial, there is speculation about the potential integration of cryptocurrency trading on Twitter's rebranded platform, X. Musk's history with Dogecoin and his interest in cryptocurrencies continue to be key points of discussion.	
Question	Answer	Fact answer
Who will launch their own stock and bitcoin crypto trading platform?	Elon Musk and his platform X.	Based on the given facts, it can be concluded that Elon Musk denied launching both a stock trading platform and a cryptocurrency trading platform. Therefore, no specific type of trading platform will be launched.
What will Elon Musk launch?	Stock and bitcoin crypto trading platform.	The given facts do not mention the name of Elon Musk's platform.
What is the name of Elon Musk's platform?	X.	Elon Musk will not launch a stock trading and cryptocurrency trading platform, as he denied the claim and stated that he has no intentions of introducing crypto tokens.
What type of trading platform will be launched?	Stock and bitcoin crypto.	Based on the given facts, it can be concluded that Elon Musk's platform will not be trading any stocks or cryptocurrencies.
What will be traded on Elon Musk's platform?	Stock and bitcoin crypto.	Elon Musk will not launch his own stock and bitcoin crypto trading platform.

Fig. 5. A case study of the proposed XMDFaVer with GPT3.5 on the Misbar dataset.

need for further addition of lower-ranked reference articles into this pipeline. In addition, the experimental results indicate that the “statistical threshold classifiers,” which rely solely on manually defined median and mean values to classify news as fake or true, perform significantly worse than machine learning-based classifiers. These statistical methods use fixed thresholds for classification and do not adapt to data-specific patterns or contextual signals. As such, the relatively poor performance of these classifiers is expected and underscores their limited ability to handle the complexity of misinformation detection tasks. Their inclusion in our experiments serves as a useful baseline to highlight the clear performance gains offered by more flexible and data-driven models such as those based on machine learning and large language models.

Summarized claim	Wearing face masks and maintaining social distancing protocols will help prevent the spread of COVID-19.	
Summarized justification	A study of 172 studies in 16 countries found that masks and distancing reduce COVID-19 risk. Mask use cut risk to 3%, and six-foot distancing to 2.6%.	
Question	Answer	Fact answer
What will help prevent the spread of COVID-19?	Wearing face masks	Wearing face masks and maintaining social distancing.
How can social distancing protocols help?	Prevent spread of COVID-19	To reduce the risk of COVID-19 transmission
What measures should be taken?	Wearing face masks, maintaining social distancing	Wear face masks and follow social distancing guidelines
What is the purpose of wearing face masks?	Prevent spread of COVID-19	To lower the risk of catching COVID-19.
How can COVID-19 transmission be reduced?	Wearing masks, social distancing	COVID-19 transmission can be reduced by following guidelines that suggest wearing a face mask and maintaining social distancing protocols.

(a) A true positive example.

Summarized claim	According to social media, Tom Brady, one of the most accomplished football quarterbacks of all time, lost by 35 points against the New Orleans Saints.	
Summarized justification	Tom Brady and the Tampa Bay Buccaneers lost 38-3 to the Saints, the worst defeat of his career. Drew Brees regained the NFL touchdown record, leading 564 to Brady's 561.	
Question	Answer	Fact answer
Who lost the game?	Tom Brady	The Tampa Bay Buccaneers lost the game
What sport was the game in?	Football	Football
How many points did Tom Brady lose by?	35 points	Based on the given text, it does not provide any information about the points
Was Tom Brady successful in the game?	No	No
What is Tom Brady known for?	Football	Football

(b) A false positive example.

Fig. 6. Examples of the proposed XMDFaVer with GPT-3.5 on the Misbar dataset. The upper figure shows the generated QG/QA content for a true positive example, where the given data is true news and XMDFaVer correctly predicts it as true. The lower figure presents a false positive example, where the given data is true news, but XMDFaVer incorrectly predicts it as false.

Third, adopting QG/QA on summarized claim and fact articles while taking advantage of the current advanced LLMs such as GPT3.5 and Llama2, XMDFaVer can provide explainability of its final decision through showing how natural language answers to the same questions align or differ with reference to the news articles and reference fact articles. This is achieved by enabling the decision-making process to be checked and validated through the comparison of natural language answers to the same questions, derived respectively from the news articles and the reference fact articles. To better understand the explainability of XMDFaVer, Fig. 5 shows a case study where the model pipeline is applied to the Misbar dataset. In this case, the summarized claim is derived from an article reporting the news about Elon Musk and his platform X launching their stock and Bitcoin cryptocurrency trading platform. GPT3.5 generates 5 pairs of questions and answers, highlighted in red and yellow. For the fact-based verification phase, the summarized justification is generated from a fact article that provides evidence for the claim. GPT3.5 then generates 5 fact answers in green, corresponding to the given questions. The examples of claim answers and fact answers show a contradiction, indicating that this claim article should be treated as false.

To better understand how XMDFaVer explains its predictions, we show two examples in Fig. 6(a) and (b). Fig. 6(a) shows a true positive case where the model correctly identifies the claim “Wearing face masks and maintaining social distancing protocols will help

prevent the spread of COVID-19" as true. The justification is based on a study that reviewed 172 smaller studies from 16 countries, showing strong evidence that both masks and distancing lower the chance of getting COVID-19. The model provides accurate answers that match this evidence, demonstrating that both actions help reduce risk. It also answers related questions well, such as why masks are used and how distancing helps, by using facts from the study. This demonstrates that XMDFaVer can extract and explain different parts of the evidence clearly.

On the other hand, Fig. 6(b) shows a false positive example about Tom Brady. Here, the model wrongly labels the claim as true. The error comes from confusion between the claim and the article. The claim says Tom Brady lost by 35 points, while the article says the Tampa Bay Buccaneers lost 38–3. Even though the numbers match, the loss is linked to the team, not Brady alone. The model incorrectly blames Brady directly, revealing a weakness in distinguishing individuals from groups. Also, when asked for exact numbers (like how many points were lost), the model doesn't give them, which shows a problem when facts are written in slightly different ways.

Overall, these examples show that XMDFaVer works well when the differences in facts are clear and easy to match. But it can make mistakes when there are small details, like who is responsible or exact numbers. To improve, the model could get better at telling apart names and roles, and better understand how facts relate, even when written differently. Future improvements could focus on enhancing named entity recognition and incorporating deeper semantic alignment techniques to handle such cases more robustly.

Fourth, with the claim article classification step, XMDFaVer adopts machine learning models similar to the literature. Our experiments have demonstrated that these models can effectively detect fake news while being simple and efficient. These models, combined with larger and more complex current LLMs for QG/QA capabilities, have been shown to be the best-performing models on the two datasets in both binary and multi-class classification settings in our experiments.

Lastly, to evaluate the inference time of XMDFaVer, we measured its computational time over five rounds, each consisting of 10 randomly selected claim samples. On average, the total processing time per claim is approximately 13,045.56 milliseconds. The majority of this time is attributed to the LLM operations for QG and QA, which take around 12,045 milliseconds per instance. In contrast, the document retrieval component is highly efficient, requiring only 0.56 milliseconds per claim on average. These results suggest that while LLM-based modules dominate the inference time, the information retrieval step remains lightweight and scalable, making XMDFaVer a feasible solution for practical fact-based misinformation detection with a strong balance between explainability and computational performance.

There are several potential ways in which this work can be further work in the online misinformation detection domain. The newly collected and created Misbar dataset provides a greater opportunity to establish factually verifiable misinformation detection. As previously reported, there is still a need to have more data in the field [37,38]. Moreover, considering that human annotation is time-consuming and expensive, future research could explore fact-verifiable online misinformation detection from the angle of unsupervised or zero-shot learning using LLMs. In addition, although XMDFaVer adopts QG/QA to achieve the interpretability of online misinformation detection, a QG/QA method based on summarized news articles is a lossy procedure where some information from within the original articles may get lost in the process. Future research could explore information loss [39,40] and its impact on the decision-making process for online misinformation detection.

7. Conclusion

Detecting misinformation through automatic fact-based verification has been proven to be a multifaceted and dynamic challenge. To overcome some challenges in the domain, a novel dataset named Misbar is presented to address the shortcomings of data to train fact-based verification models for misinformation detection, providing additional evaluation scope for online misinformation detection in long news articles. Then, considering the more general scenario in the real world of misinformation detection on long documents, a new online misinformation detection approach is proposed utilizing automatic fact-based news verification, that uses an LLM-based question and answer generation mechanism that distills the article claim, employs information retrieval, and identifies factual evidence that is used by a combination of classifiers to decide whether the claimed article is true. A text summarization model in the article analysis stage is employed to address the challenge of the different and large input lengths for learning models. Several baseline models, encompassing standard baselines and those integrating top fact articles, are used for comparison. Our experiments have consistently demonstrated that the proposed approach, featuring sets with varying quantities of top fact articles, exhibits superiority over baseline models in most of the test cases. Notably, the proposed methods utilizing GPT3.5 and Gradient Boosting with the top 2 articles can outperform the baselines and the Llama2 methods in detecting misinformation. Moreover, an increased number of factual articles can result in an improvement in misinformation detection.

There are several potential ways in which this work can be further work in online misinformation detection. The newly collected Misbar dataset presents a valuable opportunity to advance fact-based misinformation detection, especially given the ongoing need for high-quality datasets in this field [37,38]. Since human annotation remains time-consuming and costly, future research could investigate unsupervised or zero-shot learning approaches using large language models (LLMs) to enable scalable and cost-effective detection.

Although XMDFaVer uses question generation and answering (QG/QA) to support interpretability, the reliance on summarized versions of news articles introduces potential information loss, which may affect the accuracy and reliability of decisions [39,40]. Future studies could analyze the extent of this information loss and its influence on misinformation detection outcomes. A promising direction for future research is to conduct comparative experiments between TF-IDF with cosine similarity and more advanced semantic retrieval techniques, such as BERT or SBERT, which may further improve retrieval accuracy and system robustness. Moreover, while this paper focuses on misinformation detection in English, the Misbar dataset is also available in Arabic. Future work should

therefore explore the effectiveness of semantic retrieval across languages, examine the model's performance in multilingual settings (including challenges related to tokenization and embeddings), and evaluate its generalizability across domains—for instance, by training on political news and testing on medical misinformation.

CRediT authorship contribution statement

Ziwei Hou: Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Bahadorreza Ofoghi:** Writing – review & editing, Supervision, Methodology, Conceptualization. **John Yearwood:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ziwei Hou reports was provided by Deakin University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Statistics of curated dataset

AveriTec. is an existing dataset we curated as introduced in Section 3.1. The original AveriTec dataset consists of news articles and associated fact-checking reports curated by the authors of the AveriFact paper, covering 1999 article pairs. Each pair includes a short statement of a news article and a corresponding verification piece from a fact-checking source, including PolitiFact, Snopes, and Africa Check. In this work, we consider the more general scenario of misinformation detection on long documents. Thus, we focus on the full articles from which the claims have been sourced.

For claim articles in the dataset, Fig. A.7(a) illustrates the distribution of word counts across whole claim articles, with the x-axis representing word count ranges and the y-axis indicating the number of articles within each range. The distribution is right-skewed, indicating that the majority of articles have lower word counts, although a considerable number of articles still have higher word counts. In particular, the initial bin (0 to 499 words) exhibits the highest count at 1545 articles, whereas the remaining bins fall from 500 to 20,000 words. This indicates that many articles in the dataset fall well within the limited length of the input, making them manageable for models with this restriction. However, the presence of articles that extend up to 20,000 words highlights a significant issue. For these longer articles, the fixed token limit necessitates summarization.

Fig. A.7(b) illustrates the distribution of character counts across whole claim articles, with the x-axis representing character count ranges and the y-axis indicating the number of articles within each range. The distribution is right-skewed, indicating that the majority of articles have 0 to 4990 character counts. Similar to Fig. A.7(a), this indicates that the claim articles need to be summarized to fit the language models.

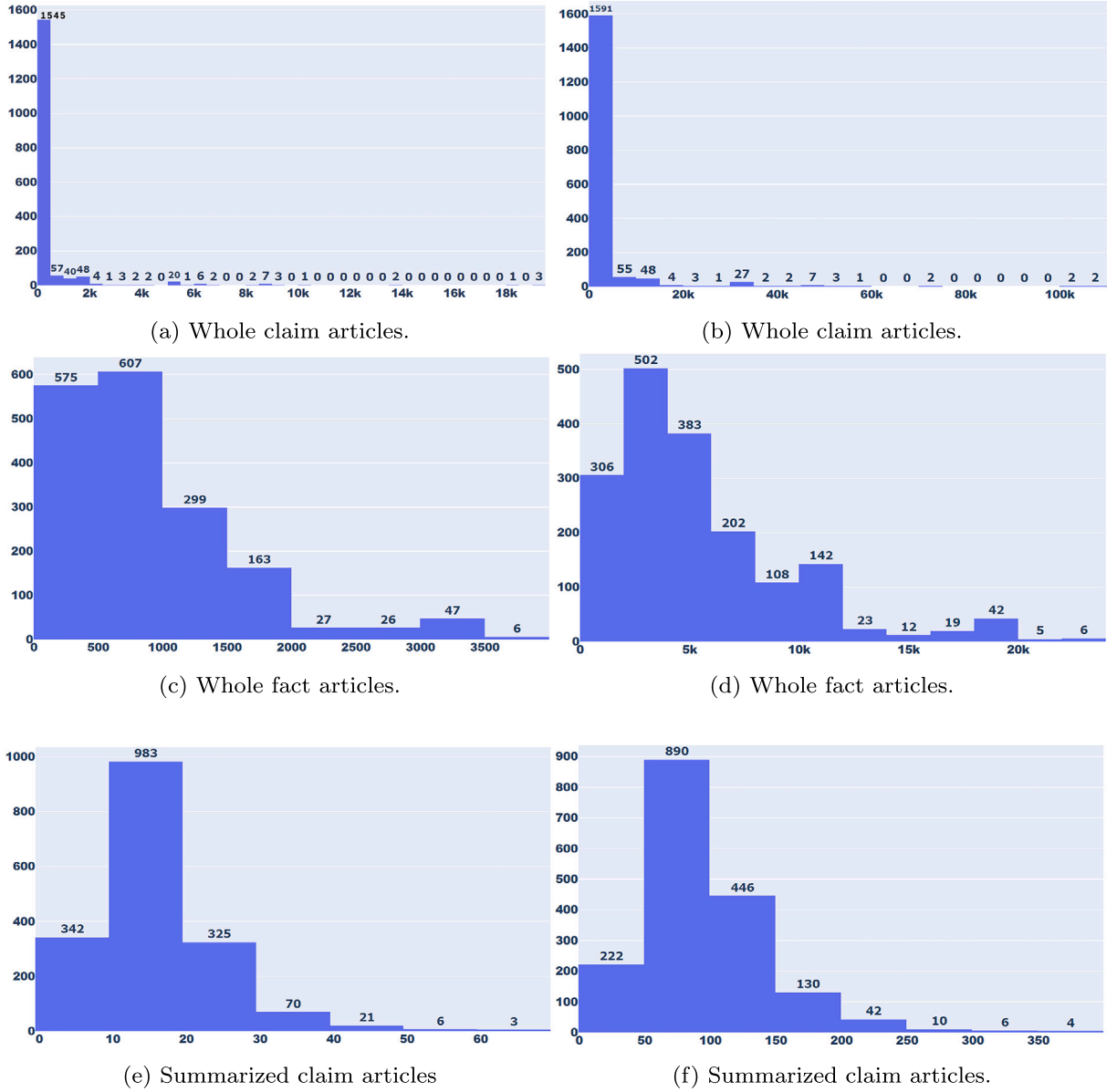
Fig. A.7(c) illustrates the distribution of document lengths within the dataset, revealing that the majority of documents are 500 to 1000 words. There are 568 documents over 1000 words in length. This pattern indicates that most documents are long documents that cannot be directly fit into the models.

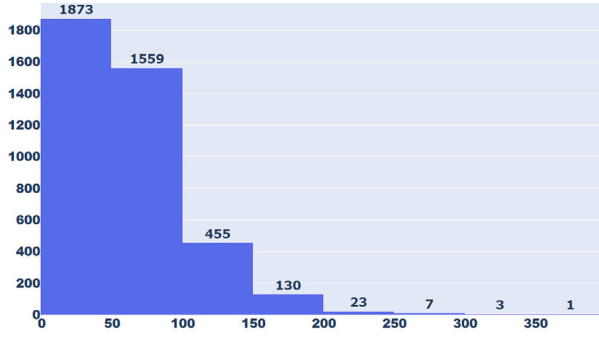
Fig. A.7(d) illustrates the distribution of character counts across whole justification articles, with the x-axis representing character count ranges and the y-axis indicating the number of articles within each range. In particular, the second bin (2000 to 3999 characters) exhibits the highest count at 502 articles. There are 942 articles containing over 4000 characters. Similar to Fig. A.7(c), it indicates that to verify the news with the facts, the fact articles need to be summarized.

For the summarized articles of the claims in the dataset, Fig. A.7(e) illustrates the distribution of word counts across a corpus of articles. In particular, the initial bin (0 to 9 words) exhibits the highest count at 342 articles. This indicates that after summarization, the length of the claim articles is significantly reduced and can fit into the language models.

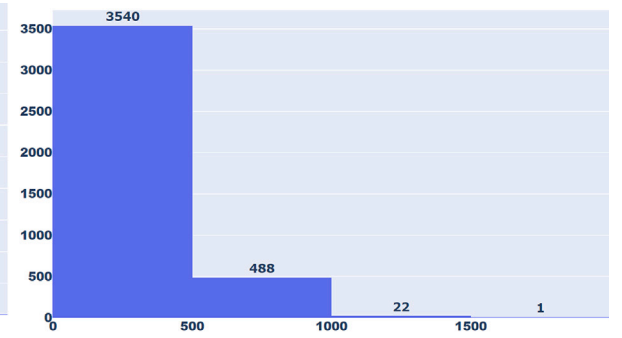
Besides, Fig. A.7(f) illustrates the distribution of character counts across a corpus of articles, with the x-axis representing character count ranges and the y-axis indicating the number of articles within each range. The distribution is notably right-skewed, indicating that the majority of articles have lower character counts, although a considerable number of articles still have higher character counts. In particular, the initial bin (0 to 49 characters) exhibits the highest count at 222 articles. This indicates that after summarization, the length of the claim articles is significantly reduced and can be fit to the language models.

Misbar. is the dataset we created to resolve the challenges of no available data in the literature. Misbar dataset is constructed by leveraging a structured web crawling process targeting the Misbar fact-checking platform which is a widely recognized and trustworthy platform in the literature [41]. Fig. A.8 illustrates the distributions of word count and characteristic count across the whole claim, fact, and summarized claim articles in the Misbar. For the whole articles of the claims in the dataset, Fig. A.8(a) illustrates the distribution of word counts across whole claim articles, with the x-axis representing word count ranges and the y-axis indicating the number of articles within each range. The distribution is right-skewed, indicating that the majority of articles have lower word counts, although a considerable number of articles still have higher word counts. In particular, the initial bin (0 to 49 words) exhibits the highest count at 1873 articles. The right-skewed distribution indicates that shorter articles, particularly those in the 100 to 300-word range, are the most common, while longer articles are still present but less frequent. This pattern indicates that most documents are long documents that cannot be directly fit into the models.

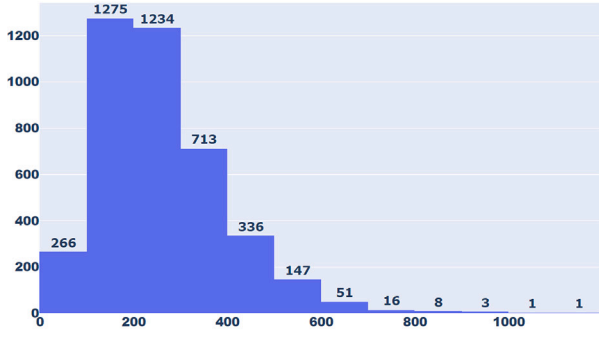




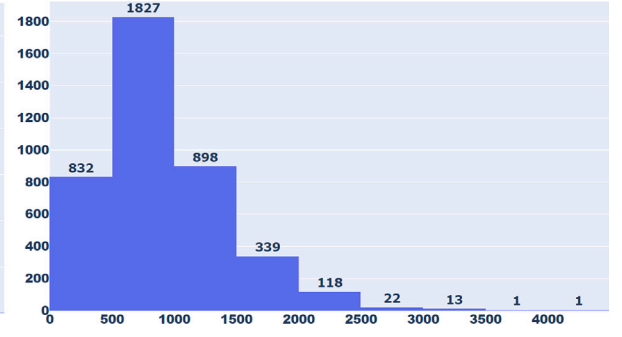
(a) Whole claim articles.



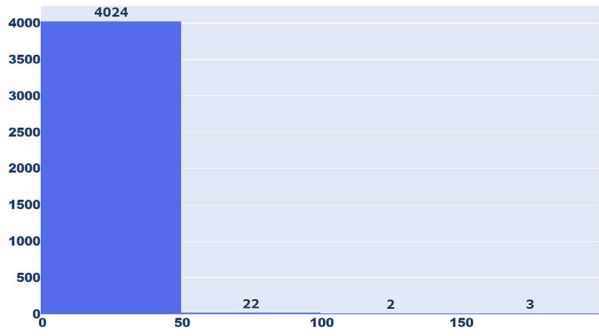
(b) Whole claim articles.



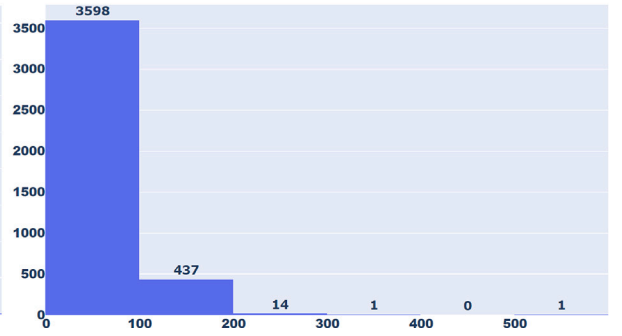
(c) Whole fact articles.



(d) Whole fact articles.



(e) Summarized claim articles



(f) Summarized claim articles.

Fig. A.8. Distributions of words and characters in the Misbar Dataset. Figures a, c, and e indicate the related distributions of word counts. Figures b, d, and f indicate the related distributions of character counts.

For the summarized articles of the claims in the dataset, Fig. A.8(e) illustrates the distribution of word counts across a corpus of articles, with the x-axis representing word count ranges and the y-axis indicating the number of articles within each range. This indicates that after summarization, the length of the fact articles, measured in words, is significantly reduced, making them manageable within the input constraints of language models.

Besides, Fig. A.8(f) illustrates the distribution of character counts across a corpus of articles, with the x-axis representing character count ranges and the y-axis indicating the number of articles within each range. Similar to Fig. A.8(e), this indicates that after summarization, the length of the claim articles is significantly reduced, allowing them to be processed effectively by language models.

Appendix B. Fact retrieval-semantic-based retrieval

To further evaluate our proposed retrieval approach, we conducted a comparative experiment using a semantic retrieval model based on Sentence-BERT (SBERT) [42]. Specifically, we employed the publicly available all-MiniLM-L6-v2 model, which encodes textual inputs into dense embeddings and computes semantic similarity via cosine similarity. This method has been widely adopted in natural language processing tasks involving semantic matching and retrieval.

Table B.9

Fact retrieval performance on Extended-AveriTec and Misbar datasets using sBERT. Note: Accuracy@T1 and Accuracy@T2 are the accuracies of successfully finding the corresponding fact articles in the top 1 and top 2 fact articles and MRR@1 and MRR@2 are the mean reciprocal ranks (MRR) at the top 1 and top 2 retrieved articles.

Dataset	Accuracy@T1	Accuracy@T2	MRR@1	MRR@2
Extended-AveriTec	44.31 %	50.00 %	0.4431	0.4716
Misbar	14.94 %	29.86 %	0.1494	0.2240

The retrieval task was evaluated under the same experimental setup as our proposed method based on TF-IDF and cosine similarity. We report retrieval accuracy at T1, T2, MRR@1, and MRR@2 (Table B.9).

The results show that, in this setting, TF-IDF with cosine similarity performs much better than SBERT. This is mainly due to the characteristics of the task. Claims are usually short and often include very specific words such as names, dates, numbers, or organizations. These uncommon terms are especially useful for matching claims with evidence, and TF-IDF naturally gives them higher weight. By focusing on exact word overlap, TF-IDF is able to effectively capture these distinctive terms, which are critical for precise fact retrieval.

In comparison, SBERT is designed to measure overall semantic similarity between pieces of text. While this works well for tasks like paraphrase detection, it does not always align with factual correctness. Two sentences can be semantically close but still communicate opposite meanings—for example, one may support a claim while the other challenges it. This is a common situation in fact-checking, where evidence may either agree with or contradict a claim. Because SBERT tends to emphasize surface-level similarity, it struggles to capture this distinction. Furthermore, by compressing an entire article into a single dense vector, SBERT can lose some of the fine-grained lexical cues that are critical for accurate retrieval.

These factors help explain why SBERT performs worse in this setting. TF-IDF, by contrast, preserves important lexical signals and directly matches claims with relevant evidence, making it more reliable for claim–fact retrieval.

Overall, these findings support the use of TF-IDF with cosine similarity as a solid and interpretable retrieval method for our system. At the same time, they highlight room for future research. In particular, domain-adapted or fine-tuned semantic models could be explored to better distinguish between supporting and opposing evidence while still benefiting from the flexibility of neural embeddings. A hybrid approach may combine the interpretability of TF-IDF with the broader generalization strengths of embedding-based methods.

Data availability

Data will be made available on request.

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